NAME:…………………………………………………….…….. ADM …..….….… STREAM:…………....

DATE: ....

121/2
MATHEMATICS
PAPER 2
TIME: 2 ½ HOURS

IKUU BOYS APRIL HOLIDAY ASSIGNMENT 2026

FORM THREE MATHEMATICS PAPER 2

INSTRUCTIONS TO CANDIDATES

- Write your name and Admission number in the spaces provided at the top of this page.
- This paper consists of two sections: Section A and Section B.
- Answer ALL questions in section A and ONLY FIVE questions from section B.
- All answers and workings must be written on the question paper in the spaces provided below each question.
- Show all the steps in your calculation, giving your answer at each stage in the spaces below each question.
- Non – Programmable silent electronic calculators and KNEC mathematical tables may be used, except where stated otherwise.

FOR EXAMINERS USE ONLY

SECTION A

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

SECTION B

17	18	19	20	21	22	23	24	TOTAL

GRAND TOTAL

This paper consists of 15 printed pages. Candidates should check the question paper to ascertain that all pages are printed as indicated and that no pages are missing.

SECTION A (50 MARKS) (*Answer all questions in this section*)

1. By completing the square, solve for value x in the equation $2x^2 - 6 = x$ (3 marks)

2. Use logarithm to evaluate: $\sqrt[3]{\frac{0.08294 \times 39.24}{8\,458}}$

(3 marks)

3. Find all the integral values of x which satisfy the inequality. (3 marks)

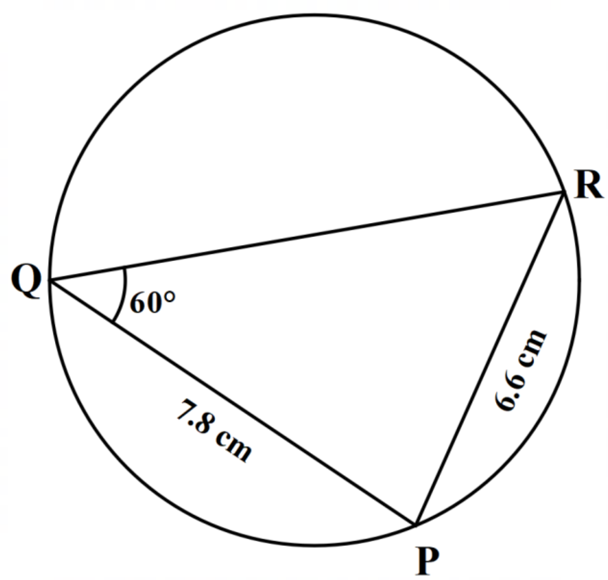
$$3(1+x) < 5x - 11 < x + 45$$

4. Without using tables or calculator, solve for value of x.
(3 marks)

$$\frac{1}{2}\log_3 729 - 2\log_3 x + 1 = 0$$

5. Find the percentage error in calculating the volume of the cuboid whose dimensions are 8.2cm by 6.2cm by 5.7cm.
(3 marks)

6. Triangle PQR is circumscribed in the circle as shown. PQ = 7.8cm, PR = 6.6cm and angle PQR = 60°. Calculate the radius and circumference of circumcircle of the triangle.
(3 marks)

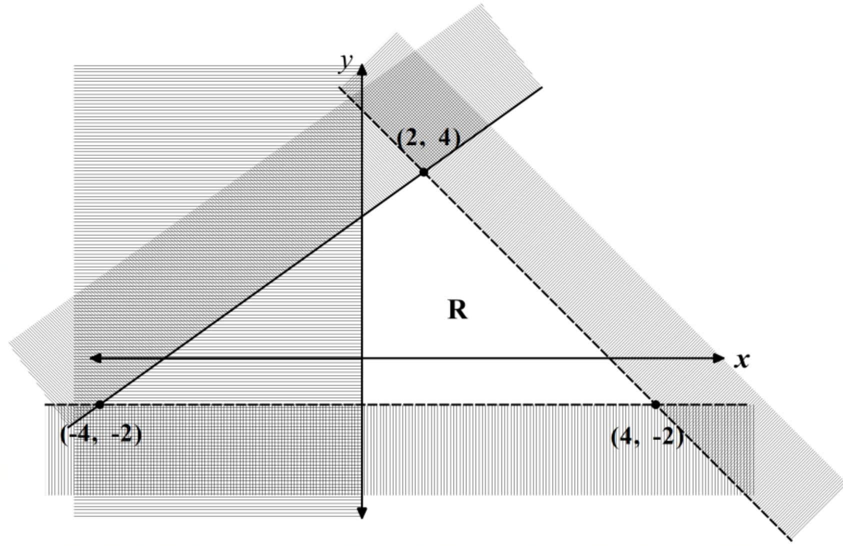


7. Simply and leave answer in surd form.
(3 marks)

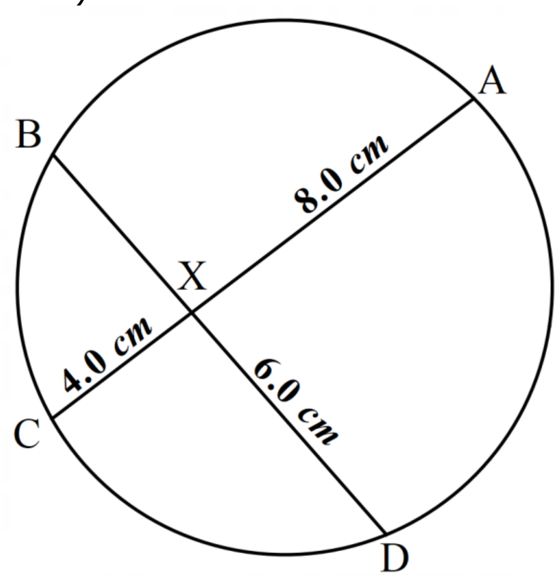
$$\frac{-9}{\sqrt{13}+\sqrt{3}} + \frac{5}{\sqrt{13}-\sqrt{3}}$$

8. Eunice invested Sh. 6400 at 15% per annum compound interest for 3 years. Azanat invested twice that amount at 12.5% per annum simple interest for the same period of time. Find whose investment earned more interest and by how much. (4 marks)

9. Write down the four inequalities which define the region R.
(4 marks)



10. In the diagram below, X is the point of intersection of the chords AC and BD of the circle such that AX = 8cm, XC = 4cm and XD = 6cm. Find the length of XB. (3 marks)

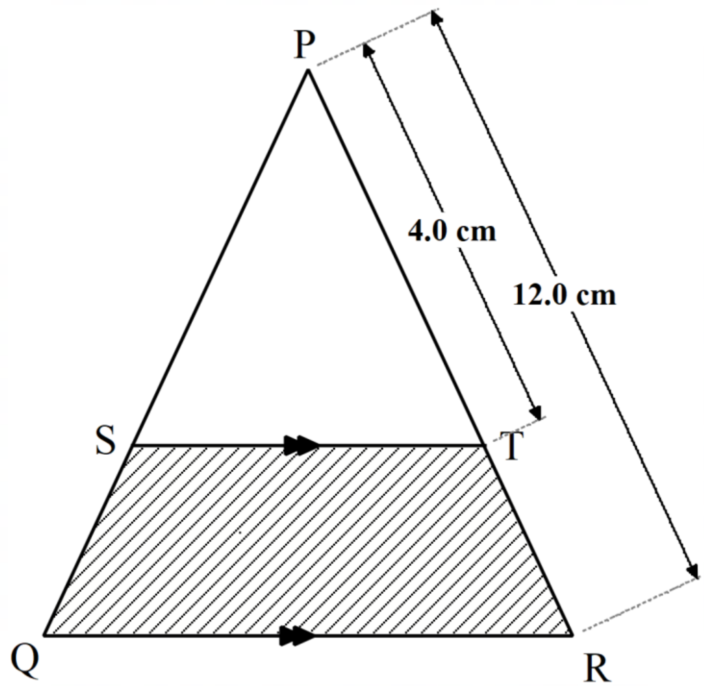


11. Using reciprocal tables only evaluate $\frac{3}{0.01492} + \frac{12}{16.58}$ correct to 4 significant figures.
(3 marks)

12. Madam Akinyi earns a basic salary of Ksh. 24,000 per month. In addition, she is paid a commission of 5% for sales above Ksh. 30,000. In the month of February, she sold goods worth Ksh. 300,000 at a discount of 6%. Calculate her total earning that month.
(3 marks)

13. A two-digit number is such that the sum of the digits is 11. When the digits are interchanged the new number formed is 45 less than the original number. Determine the original number.
(3 marks)

14. The figure below shows triangle PQR in which PR = 12cm. T is a point on PR such that TR = 4cm. Line ST is parallel to QR. If the area of PQR is 336cm2. Find the area of the quadrilateral QSTR.
(4 marks)



15. The number $5.\dot{8}\dot{1}$ contains an integral part and a recurring decimal. Convert the number into a mixed number
(2 marks)

16. Francis bought a vehicle at Ksh. 2 800 000. After three years he sold the vehicle at Ksh. 1,500,000. Determine the average rate of depreciation per annum correct to one decimal place.
(3 marks)

<u>SECTION B (50 MARKS) (*Answer only five questions in this section.*)</u>

17. The table below shows income tax rates for a certain year.

Monthly income in Kenya shilling (Ksh)	**Tax rate in each shilling**
0 - 10164	**10%**
10165 - 19740	**15%**
19741 - 29316	**20%**
29317 - 38892	**25%**
Over 38892	**30%**

A secondary school teacher was earning a monthly basic salary of Ksh. 55,480 house allowance of Ksh. 12,000 and a commuter allowance of Ksh. 8000. He was entitled to a personal relief of Ksh. 1162 per month.

a) Calculate

I. The teacher's taxable income (2 marks)

II. The teacher's net monthly tax (5 marks)

b) In addition to the tax the other deductions were per month as follows: (3 marks)

- Cooperative loan Ksh. 10,000
- Co-operative shares Ksh. 2,000
- Window and children's pensions scheme at % of the basic salary.

Calculate the teacher's net monthly pay.

18. A certain number of people agreed to contribute equally to buy books worth shs.1200 for a school library. Five people pulled out and so that others agreed to contribute an extra sh.40 each. Their contribution enabled them to raise the sh.1200 expected.

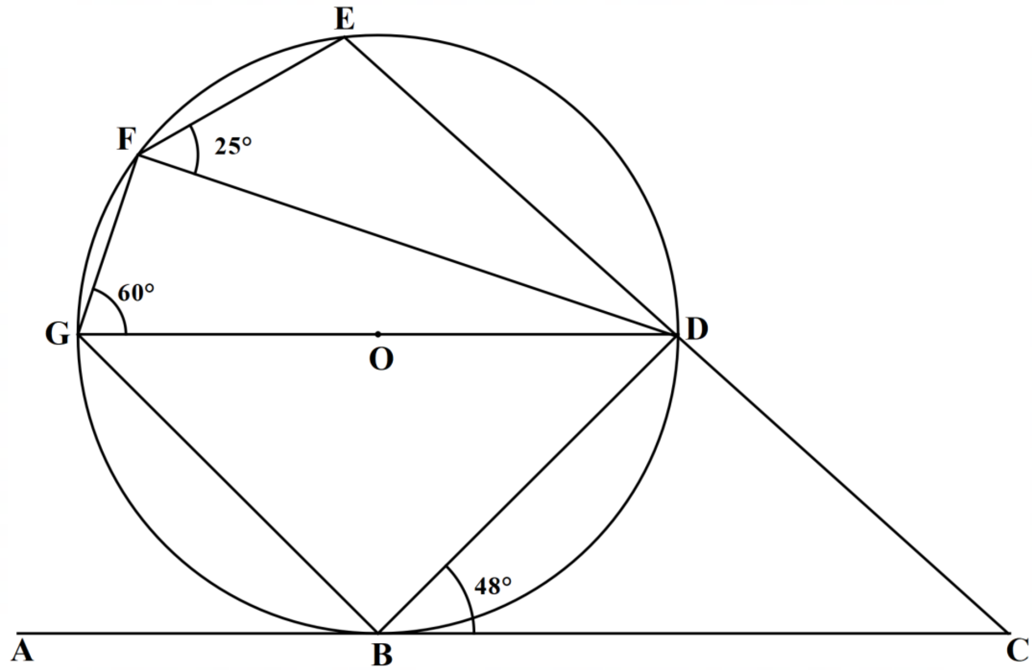
a) If the original number of people was x, write an expression of how much each was originally going to contribute.
(1 mark)

b) Write down two expressions representing how much each contributed after the five people pulled out.
(2 marks)

c) Calculate how many people made the contribution.
(5 marks)

d) Calculate the amount made by each of the remaining members.
(2 marks)

19. In the figure below, ABC is a tangent to the circle, centre O. DOG is a diameter and angle DGF = 60°, angle DBC = 48° and angle DFE = 25°.



Giving reasons, find the size of angles:

a) FED (2 marks)

b) Obtuse FOB (2 marks)

c) EBD (2 marks)

d) BCD (2 marks)

e) OBE (2 marks)

20. The frequency table shows masses, to the nearest kilogram of 60 fish caught by a fisherman in a day.

Mass (kg)	**5-9**	**10-14**	**15-19**	**20-24**	**25-29**	**30-34**	**35-39**	**40-44**
No: of fish	**6**	**x**	**12**	**10**	**5**	**6**	**2**	**1**

a) Find the value of x.
(2 marks)

b) Draw a histogram to represent the above information.
(5 marks)

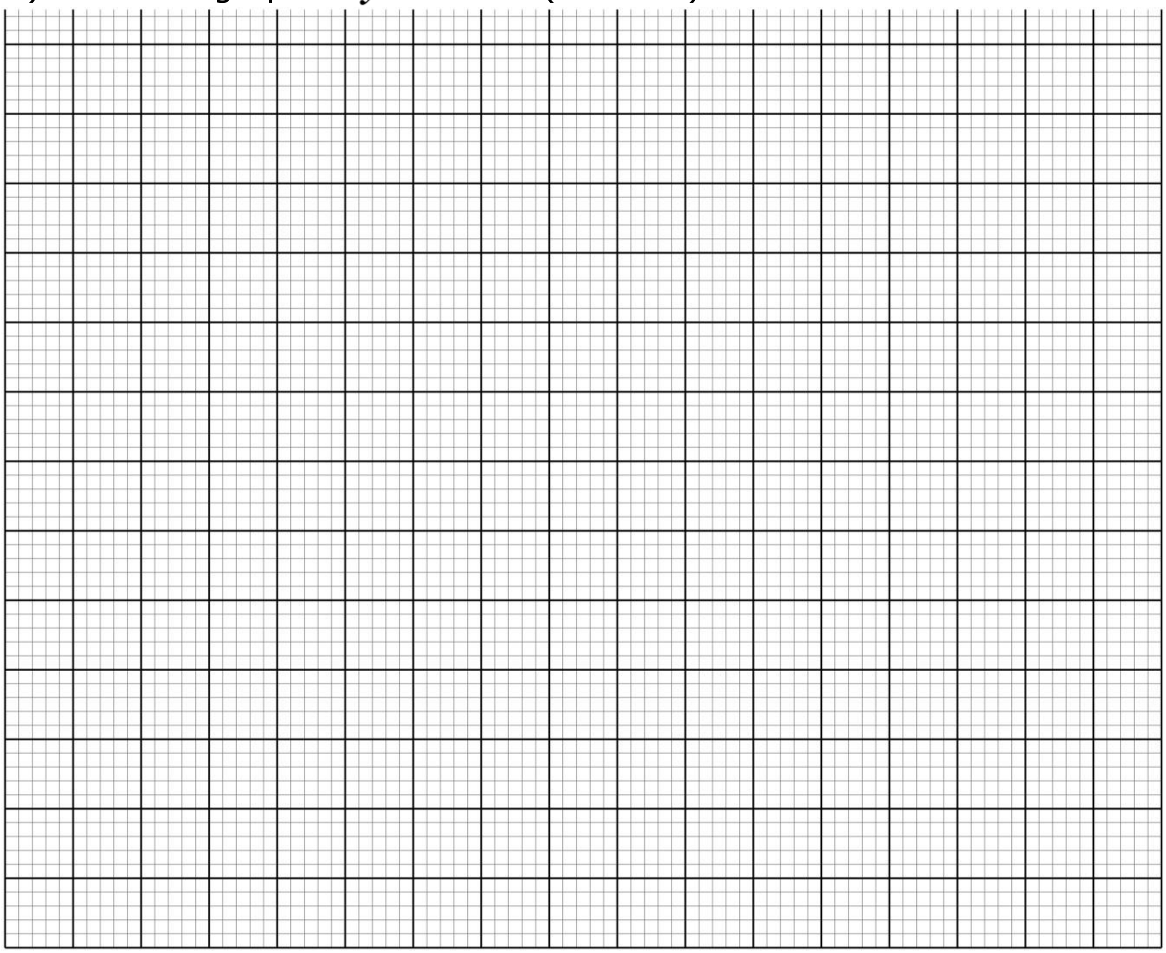
c) (i) State the class in which the median mass lies.
(1 mark)

(ii) Draw a vertical line in the histogram, showing where the median mass lies.(2 marks)

21. a) Complete the table below for the function $y = Sin\,x$.
(2 marks)

X°	0	30	60	90	120	150	180	210	240	270	300	330	360
$y = Sin\,x$	**0**		**0.87**					**-0.5**					**0**

b) Draw the graph of $y = Sin\,x$.(3 marks)



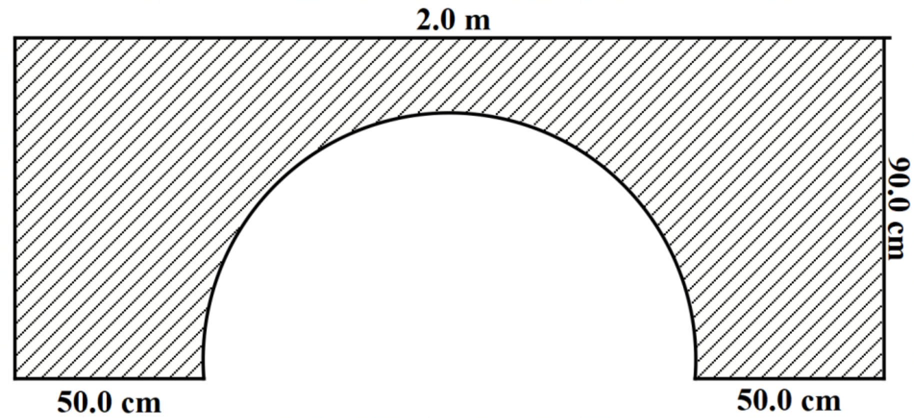
c) Use the graph to solve the following equations:

I. **Sin X = 0.25**
(2 marks)

II. **Sin X = -0.60**
(2 marks)

III. **Sin 70 =?**
(1 mark)

22. The diagram below shows the cross-section of a culvert of length 12m.



a) Calculate the:

I. Cross-sectional area of the culvert. (3 marks)

II. The volume of the concrete that was used to make the culvert. (2 marks)

b) The mixture of cement, sand and gravel was used to prepare concrete of density 2.4 g/cm3. If the ratio of cement, sand and gravel is 1:3:4, calculate the number of bags of cement used given the mass of one bag of cement is 50kg. (5 marks)

23. A straight line $\mathbf{L_1}$ has a gradient of **-2** and passes through the point **P (-1, -3)**.

Another line L_2 passes through the points **Q (1, -3)** and **R (4, 5)**. Find:

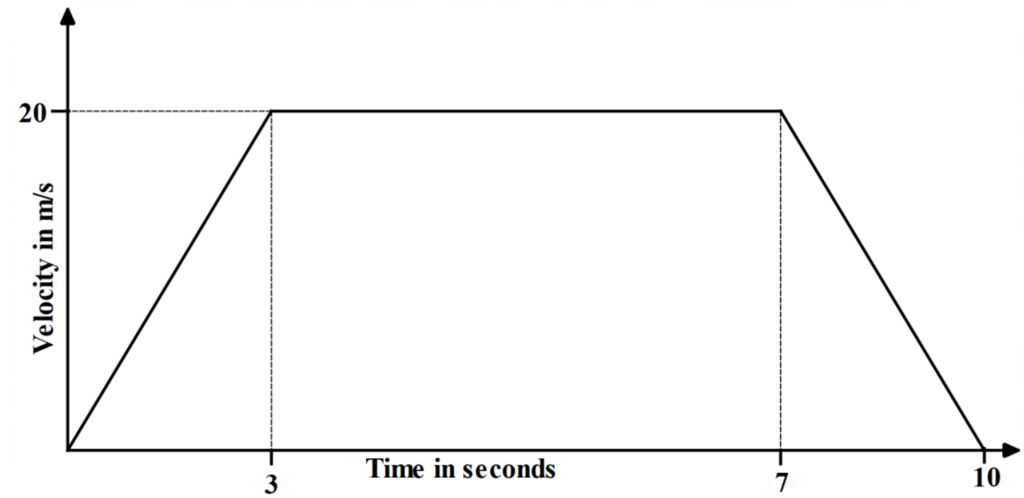
a) The equation of $\mathbf{L_1}$.
(2 marks)

b) The equation of line $\mathbf{L_2}$.
(3 marks)

c) The point of intersection of lines $\mathbf{L_1}$ and $\mathbf{L_2}$.
(3 marks)

d) The equation of line $\mathbf{L_3}$ passing through point **R (4, 5)** and parallel to $\mathbf{L_1}$. (2 mar

24. The velocity-time graph below shows the motion of the car for 10 seconds.



Use this velocity - time graph to find:

a) The rate of acceleration.
(2 marks)

b) The rate of retardation.
(2 marks)

c) The total distance travelled.
(2 marks)

d) The average speed maintained during this journey.
(2 marks)

e) The distance travelled at the constant speed.
(2 marks)